

COMP1004 Project Portfolio 2025 2026

Kayleigh Soane - University of Plymouth
Created: 09-OCT-25 Last Modified: 02-JUN-26



Contents

[COMP1004 Project Portfolio](#)

[Portfolio Information](#)

[Indicative Scenarios](#)

[Allowed Technologies List](#)

[1 Project Initiation](#)

[2 Mid Project Review](#)

[2.1 Video Submission](#)

[2.2 Professional Development Submission](#)

[3 Portfolio Submission](#)

[3 Final Portfolio](#)

[3.1 Final Professional Development Reflection](#)

[3.2 Final Project Plan Submission](#)

[3.3 UML Diagram Submission](#)

[3.4 GitHub Repository Link](#)

[3.5 Poster Submission](#)

[3.6 Project Reflective Account](#)

[3.7 Completed AI and Marketing Declaration](#)

[4 Viva](#)

COMP1004 Project Portfolio

Kayleigh Soane - University of Plymouth

Created: 09-OCT-25 **Last Modified:** 25-NOV-25



Contents

[Portfolio Information](#)

[Indicative Scenarios](#)

[Allowed Technologies List](#)

Portfolio Information

Welcome to the COMP1004 Project Portfolio

This is the first of three project modules you will undertake during the course of your degree. This is a 40 credit, all year module, going over the two Semesters, where you carry out an **individual self-led** piece of work creating a Single Page Application (SPA) using HTML, CSS and JavaScript.

It is recommended that you work on a project suitable for your degree where you will focus on the early stages of the software development lifecycle (SDLC) whilst exploring the operational and technical trade-offs that are inherent in that cycle. Indicative scenarios can be found either in the "Indicative Scenarios" section. Each scenario has an indicated base functionality and an initial task is to research the background to the project and to develop that functionality further.

Of course you are completely free to choose your own topic area/problem statement. The project will be run using the Agile Project Management. As part of your portfolio you will be expected to submit interim deliverables throughout the duration of the project as well as demonstrate something during the scheduled stand-up meetings every 2 weeks.

You are free to select any theme you like. However, all projects must demonstrate the following:

- An index HTML page
- JSON output to a flat file
- JSON input from a flat file
- A single-page application showing some form of interactivity.

Although you are free to select any theme you like, some project ideas with the theme 'self-management' have been provided. Consider time in terms of work-life balance and how to achieve this. The project scenarios provided are the minimum specifications. It is up to you to develop these further and add further functionality. Applications that can demonstrate novelty and self-led learning will achieve extra marks.

You **must only** use technologies that you have learnt during this module.

Important Dates to Note

Please refer to the table below for the submission dates for COMP1004.

	Submission Deadline	Feedback
Interim Video and Professional Development Review	Thursday 15th January 2026 at 15:00	Within 20 working days
Final Portfolio	Thursday 30th April 2026 at 15:00	Within 20 working days
Viva	Tuesday 12th May	Within 20 working days

Indicative Scenarios

Indicative Scenarios

This page outlines some indicative scenarios that you may wish to explore. They are all based around the theme of time management.

Games Development Technologies

The focus of this project is to explore leisure time and consider what might enrich this. Additionally, the project should also record time spent playing the game and output this into a flat file.

Produce a hand coded hypertext fiction game using HTML, CSS and JavaScript enhancements. Your game must be playable from start to finish. You will need some custom art for the game pages. There should be a clear narrative structure and plot. The game structure should be coded in a clear, logical manner using JavaScript.

Your application should output to a flat file the following JSON data format. It should have a player username, minutes the player has spent playing, their score and the level.

Cyber Security

The focus of this project is to provide a password account management system for a user accessed via a browser. Produce a hand coded application whereby an individual can manage their own passwords and accounts for different applications. As a minimum your application should provide the following functionality:

- Use local storage to store personal account data such as username and password.
- Read a JSON file using the format above to identify a matching username in your system.
- Indicate password strength to the end user.

You will need to determine what additional features this application should have and the best way to visualise this information for the user.

Computer Science

The focus of this project is to create an algorithm and information page to allow an end user to manage their time. As a minimum your application should provide the following functionality:

- Read a JSON file to identify how much time a player has spent on leisure activities.
- Use an appropriate visual indicator to the user in relation to time spent. Eg: Red indicates too much time on a game, green is ok. This will require you to decide what constitutes “too much time”.
- Output JSON to a new file to save information about how the user wishes to manage their time and deadlines.

You will need to determine what additional features this application should have.

Software Engineering

The focus of this project is to create an application to help manage time and provide support. Your project should explore how you might use an Alexa Skill for notifying the user via a Voice interface about their deadlines or appointments. Your application must as a minimum provide the following:

- Read a JSON file to identify the time a player has spent on a particular activity.
- Display on the web page a time management plan.
- Alert the user to an upcoming event.

You will need to determine what additional features this application should have.

Artificial Intelligence

The focus of this project is to create an application that visualises the time spent across different activities and tracks goals. As a minimum your application should provide the following functionality:

- Read a JSON file to identify the time spent by the user across different activities.
- Display how the user has spent their time graphically.
- Congratulate the user on meeting a goal.

You will need to determine what additional features this application should have.

Allowed Technologies List

Allowed Technologies List

While there are many technologies available, we ask that you use only the ones specified below for consistency and compatibility. Please refrain from using alternatives, even if they are available.

Development Environments

Microsoft Visual Studio 2022

Microsoft Visual Studio Code

Notepad++

Notepad

CSS

Bootstrap

JavaScript

Node.js – for importing your JavaScript.

AI Assistant

Microsoft Copilot

1 Project Initiation

Project Initiation Page

Stand-up Meeting Leader

Please select your Stand-Up Meeting Leader

Lauren Ansell

Muhammad Asad

Vivek Singh

Project Title

Wordio - Translation tool

Project Scope

The project scope outlines the initial essence of your project. As your project develops over the course of the year, this could change following feedback received during your stand-up meetings and from your peers.

It should identify why your project might be of importance and contain some background research for the project. The project scope should be approximately 250 words.

I have decided to make a translator. Translation tools are important in connecting people from all over the world to be able to build relationships and share knowledge.

Users will be able to input text in one box. The input language will be automatically detected or the user can manually select it. In a box adjacent, the user will be able to select the output language which will be displayed after the user has finished typing. The output will be displayed as grammatically and contextually correct in that language. The translation will be able to be copied to the clipboard.

After doing some background research, I learnt that translation software uses several types of algorithms to translate words. There are used to process factors such as grammar and to identify its meaning so that the output text is printed as grammatically and contextually correct. Some use AI to do this. I have found that I need to use a translation API as a word library that can be fetched to display the translation. These are freely accessible online, therefore my project would need to be able to access the internet.

When considering the accessibility of the website, I could implement light and dark themes as well as themes accessible to people with dyslexia or colour blindness. It would also be a good idea for the site to be readable in different languages, so I could have a language selector for the whole website similar to the theme selector.

Word count: 248 (Max 275)

Requirements

Outline 3 to 5 initial requirements for your project. These can be both functional and non-functional and listed as bullet points. As with your project scope, these could change over the course of your project.

- Input field that is processed and used for the output field
- Language filters
- Can identify inputted language without special symbols
- At least two or three languages that can be translated
- Theme and language selector
- Text to speech (potential extra)

Word count: 39 (Max 225)

Project Plan

Upload a document that outlines how your project will develop over the course of the year. Use the waterfall model to illustrate your planned activities for each sprint.

It is not possible to export '**Gantt Chart INITIAL PLAN.xlsx**' here, the file type is not supported in PDF.

2 Mid Project Review

Kayleigh Soane - University of Plymouth
Created: 26-NOV-25 Last Modified: 04-FEB-26



Contents

[2.1 Video Submission](#)

[2.2 Professional Development Submission](#)

2.1 Video Submission

Video Submission

The aim of this submission is to present and demonstrate via a video of what you have done so far and demonstrate a basic functioning prototype. The information included in your video should reflect the information recorded in section 1: Project Initiation.

It is recommended that you show your video to your peers and get feedback prior to submission.

You have 5 minutes and must present the following in this exact order:

1. The project vision and background
2. Your project plan and sprints so far
3. UML models so far
4. An initial prototype running
5. Issues/challenges faced

You can use Panopto, Zoom or Teams to record your video if needed.

Video Submission Point

Please upload your video here.

It is not possible to export '**Video Project.mp4**' here, the file type is not supported in PDF.

2.2 Professional Development Submission

Professional Development Review

As part of the series of COMP1004 lectures, you have had sessions regarding Careers. You may have also attended additional sessions hosted by university research groups.

You need to write a 250-word review reflecting on these continuous professional development (CPD) events and how they have impacted or influenced your approach to your project.

First CDP Reflection

The CPD events have taught me key things about what I need to do in order to gain a professional career. I particularly took interest in the talk with Luke. L from Microsoft, who not only explained the importance of self-promotion, but of taking advantage of any resources and opportunities I can that would help expand my skills and knowledge to try to boost me ahead of the competition. The talk was also inspiring because it gave me a glance of where I could be in five years time if I work hard. I took value in other talks too - ones that gave a little insight on how to begin developing a professional profile, where to look for opportunities, and how to improve my skills. The skills analysis session helped me identify my strengths and weaknesses and showed me what skills I needed to focus on developing. Some of my personal weaknesses were communication and programming. I am continuously developing my programming skills through my project work but also through independently seeking more information from additional reading sources and courses online, which have been helpful in improving my project too. I am steadily improving my communication skills by opening up more to new people and being more confident at sharing my ideas and opinions with others, which will be useful for me when it comes to discussing and presenting my project to others. These skills aren't only important for the development of my project, but for my professional and personal development too. Therefore I can say I have taken great influence from these sessions and they have helped me greatly.

Word count: 270 (Max 275)

3 Portfolio Submission

Kayleigh Soane - University of Plymouth
Created: 26-NOV-25 Last Modified: 19-APR-26



Contents

[3 Final Portfolio](#)

[3.1 Final Professional Development Reflection](#)

[3.2 Final Project Plan Submission](#)

[3.3 UML Diagram Submission](#)

[3.4 GitHub Repository Link](#)

[3.5 Poster Submission](#)

[3.6 Project Reflective Account](#)

[3.7 Completed AI and Marketing Declaration](#)

3 Final Portfolio

Portfolio Submission

In this section, you will submit the elements that comprise your final application.

Your final submission should contain the following items:

- A reflective account of approximately 500 words considering the issues and challenges arising from the project.
- A report of your final project plan with a maximum word count of 3000. This should illustrate your backlog and sprints.
- A complete set of UML diagrams to represent your application.
- A link to the completed and finished GitHub repo. The README file must outline your project.
- A poster illustrating the key features of your application and the architecture.

A number of previous student's portfolios have also been uploaded to the assessments tabs - these are provided to give you a better understanding of what needs to be contained within each of the elements above **NOT** best practice.

3.1 Final Professional Development Reflection

Professional Development Reflection

Now that your COMP1004 project is complete, write a 250-word reflection on a CPD activity you engaged with over the course of the year—such as (but not limited to) guest lectures run by university research groups, LinkedIn learning, online webinars/seminars.

Consider how these experiences influenced your approach to the project, shaped your skills, or supported your academic and professional development.

Second CPD Reflection

I have been interacting with the Enhance Your Future Programme. It has been a useful tool to allow me to start thinking and preparing for my future, building my skills and learning how to maximise my opportunities. So far, I have been completing the *Who Am I* section, developing my self-awareness. It has allowed me to reflect on my personality and values, my strengths and weaknesses, and identify my personal goals and motivations. This has been useful for not only my personal development, but also in finding what I want out of my career and my future. Self-reflection has allowed me to gain a clearer understanding of myself, of how I think and how my beliefs and values impact my decisions. Furthermore, it has helped me identify how I can improve myself. Through the programme, I have developed a plan for self-improvement over the summer. This includes taking initiative to develop my own projects to expand my skills and knowledge, and pushing myself to build my experience and confidence by taking more opportunities and risks outside of my comfort zone. This is only the first section of the Programme, and I plan on continuing it over the summer and through to my second year, where I can learn how to network and improve my speaking and articulation skills.

Overall, I think this programme has, and will continue to, teach me invaluable lessons that will significantly help my self-development and improve my prospects.

Word count: 244 (Max 275)

3.2 Final Project Plan Submission

Portfolio Submission

Your final submission should contain a representation of your final project plan. Within this report, you should discuss the following:

- the Software Development Lifecycle and how you have applied this to your project.
- illustrate your backlog and outline,
- reflect on the work undertaken in each of the sprints.
- outline and discuss both functional and non-functional requirements and how these may have evolved or changed over the duration of your project.
- outline any Legal, Social, Ethical and Professional issues that affected or may affect your project.

A template has been provided for you under the assessments tab ("Template.pdf"). Please delete the light grey prompt text contained within this document.

A number of students' portfolios have also been uploaded to the assessments tabs - these are provided to give you a better understanding of what needs to be contained within your portfolio, **NOT** best practice.

The approximate word count is 3000+/- 10% as per University rules. You must plan your time appropriately to allow time to draft and re-draft.

It is not possible to export 'Final Project Portfolio.docx' here, the file type is not supported in PDF.

3.3 UML Diagram Submission

UML Diagram Submission

Please upload a document containing the complete set of UML diagrams to represent your final application. You may wish to include intermediate UML diagrams to illustrate how your application has developed from the project initiation to the final application.

It is not possible to export 'UML Diagrams.docx' here, the file type is not supported in PDF.

3.4 GitHub Repository Link

Link to your GitHub repository

[KayleighSoane/COMP1004ComputingProject](https://github.com/KayleighSoane/COMP1004ComputingProject)

README included

☒ A README has been provided

3.5 Poster Submission

Poster Submission

Please upload a pdf of your final poster to be displayed at your viva.

Wordio

Auto

English

What is Wordio?
An easy-to-use translation tool made with accessibility and diversity in mind

International students, making up a quarter of the student population, express difficulties integrating into UK universities due to the language barrier. This can affect people's confidence socially and academically, preventing them from reaching out to others and achieving their full potential in their studies.

Wordio was created with the intention to not only make resources to help more accessible, but also to raise awareness and remind people to look out for and help others who may be struggling. This applies not just to international students, but to everyone.

What can it do?

- Translate between any two languages
- Includes translation between Morse code and English
- Can automatically detect the language of your inputted text
- History tab keeps track of previous searches, making them accessible to recall and reinput
- Text-to-speech allows you to hear the pronunciation of each language
- Fun feature displaying a language fact for each output language

Accessibility features

It is important for a translation tool to be useable by everyone.

- Straightforward UI
- Contrasting colours
- Keyboard shortcuts
- Text-to-speech
- Compatibility with screen readers

Translate

 **UNIVERSITY OF PLYMOUTH**

Kayleigh Soane
BSc(Hons) Computer Science
kayleigh.soane@students.plymouth.ac.uk

References:
<https://www.researchprospect.com/problems-faced-by-international-students-in-the-uk/>
<https://commsnational.plymouth.ac.uk/research-briefing/2978/>
<https://boldbrains.uk/2025/01/10/overcoming-language-barriers-for-international-students-in-the-uk/>


Google Cloud

3.6 Project Reflective Account

Project Reflective Review

In this section, you have 500 words to reflect on the issues and challenges you faced over the course of developing your application.

One of the main challenges of this project was my limited experience with HTML, CSS and JavaScript. The first few stages of development took longer because I was still learning as I was developing. If I had more experience, some of these earlier features could have been reduced to less sprints and given me more time to develop some of the later features which were more complex. Furthermore, if I had a better understanding of the expectations of this project, I could have created a more effective plan from the beginning. The one I made originally lacked the level of technicality that was required, which meant I had to come up with some more ideas halfway through the development process, putting more pressure on later sprints.

Another significant setback was the confusion that occurred over whether I needed to make backend security to protect the API key. I spent a lot of time researching and attempting to create a backend proxy server, which turned out to be unnecessary and out of scope for the expectations of this assignment. I could have spent this time implementing the site language selector, a feature that would give the application more accessibility, that I originally wanted to prioritise but ultimately ran out of time to complete.

An aspect of the project I would improve if I had planned better or had more time is the design of the dropdown menus. I had implemented them using select elements, which limited the amount of styling I could apply to them. I could have written them differently to not only make them more polished, but also to add the additional feature I came up with where users could save recent selections to the top of the dropdowns. This would have supported my goal of making the site convenient to use. Similarly, I encountered an issue where the auto-detect language option affected the swap and text-to-speech functions. It is possible that if the system was written differently, I could have implemented a more reliable solution than the workaround I created that makes it default to English.

These challenges have given me a valuable learning experience that will affect how I approach future development work. They have highlighted the importance of thorough research and planning, especially when working with time constraints, and adaptability when unexpected issues arise. I have developed stronger technical skills, improved my problem-solving abilities and gained a clearer understanding of how a project should be structured from start to finish. Despite the setbacks I faced, I was able to create a functioning product that I didn't feel capable of making at the beginning of this process. If I began this project again with what I know now, I am confident that I would be able to make a more robust and thoroughly designed application.

Word count: 469 (Max 550)

3.7 Completed AI and Marketing Declaration

AI Declaration

Please upload your signed AI declaration form here.

It is not possible to export 'Student Declaration of AI Tool use in this Assessment.pdf' here, the file type is not supported in PDF.

Marketing Materials Declaration

Your project may be used as a demonstration for marketing purposes. Please confirm below your preference.

- ☒ Yes, I am happy for my submission to be used for marketing purposes
- ☐ No, I do not wish my submission to be used for marketing purposes.

4 Viva

Viva

The final part of the module is a project viva that will take a maximum of 6 minutes. During this session, you should have your application running to present to two examiners. During this time, you will be asked questions about your code, its development and the SDLC.

Within 6 minutes you must present the following, in this exact order:

- Briefly introduce your project.
- Run your application.
- Briefly go through your source code and explain how you implemented it.
- High level summary of Issues and challenges faced.
- Show your poster.

There is no portfolio submission requirements for the Viva, this is just for information.

Overall Progress	100%
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COMP1004 Project Portfolio	(100%)
Portfolio Information	(100%)
Indicative Scenarios	(100%)
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1 Project Initiation	(100%)
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